

Enhancing Operational Efficiency and Patient Experience at HealthFirst Care

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Executive Summary

- HealthFirst Care experienced increasing patient complaints related to wait times and communication delays.
- Data analysis identified peak scheduling bottlenecks in General Medicine and Cardiology.
- Resource utilization analysis showed uneven staff and equipment allocation across departments.
- Patient feedback scores were strongly

Introduction

- HealthFirst Care launched an operational improvement initiative to enhance patient experience.
- The analysis focused on reducing delays, optimizing resource usage, and improving stakeholder communication.
- Business Analysis methods were applied using BRD, RTM, stakeholder analysis, and risk assessment frameworks.
- Operational datasets were analyzed using

Business Objectives

- Reduce patient wait times by 25%.
- Improve patient satisfaction scores across all departments.
- Optimize utilization of medical resources and staff.
- Improve communication between hospital stakeholders.
- Standardize appointment scheduling workflows.
- Support leadership with dashboard-driven

Business Requirement Document (BRD)

- Problem Statement: Patients experience long wait times and inconsistent service quality.
- Key Requirement: Implement centralized scheduling and resource tracking.
- Key Requirement: Improve communication between staff and departments.
- Constraint: Limited budget and existing legacy systems.
- Constraint: Staff availability and training timelines.

Requirement Traceability Matrix (RTM)

- REQ-001: Reduce appointment delays (Must Have).
- REQ-002: Improve patient communication workflows (Should Have).
- REQ-003: Optimize resource allocation (Must Have).
- REQ-004: Implement reporting dashboards (Could Have).
- REQ-005: Enhance stakeholder visibility into operations (Should Have).

Stakeholder Analysis & Engagement Plan

- Key stakeholders include hospital executives, doctors, nurses, administrative staff, and patients.
- Executives hold high influence and require weekly progress updates.
- Doctors and nurses require workflow alignment and process training.
- Patients are primary beneficiaries and require transparent communication.
- Engagement strategies include workshops,

Scope Management Plan

- In Scope: Scheduling optimization, resource tracking, and operational reporting.
- Out of Scope: Clinical treatment changes and external vendor systems.
- Assumption: Existing hospital systems support integration requirements.
- Constraint: Limited project budget and timeline.
- WBS Phases: Requirements, Analysis, Design, Testing, Deployment.

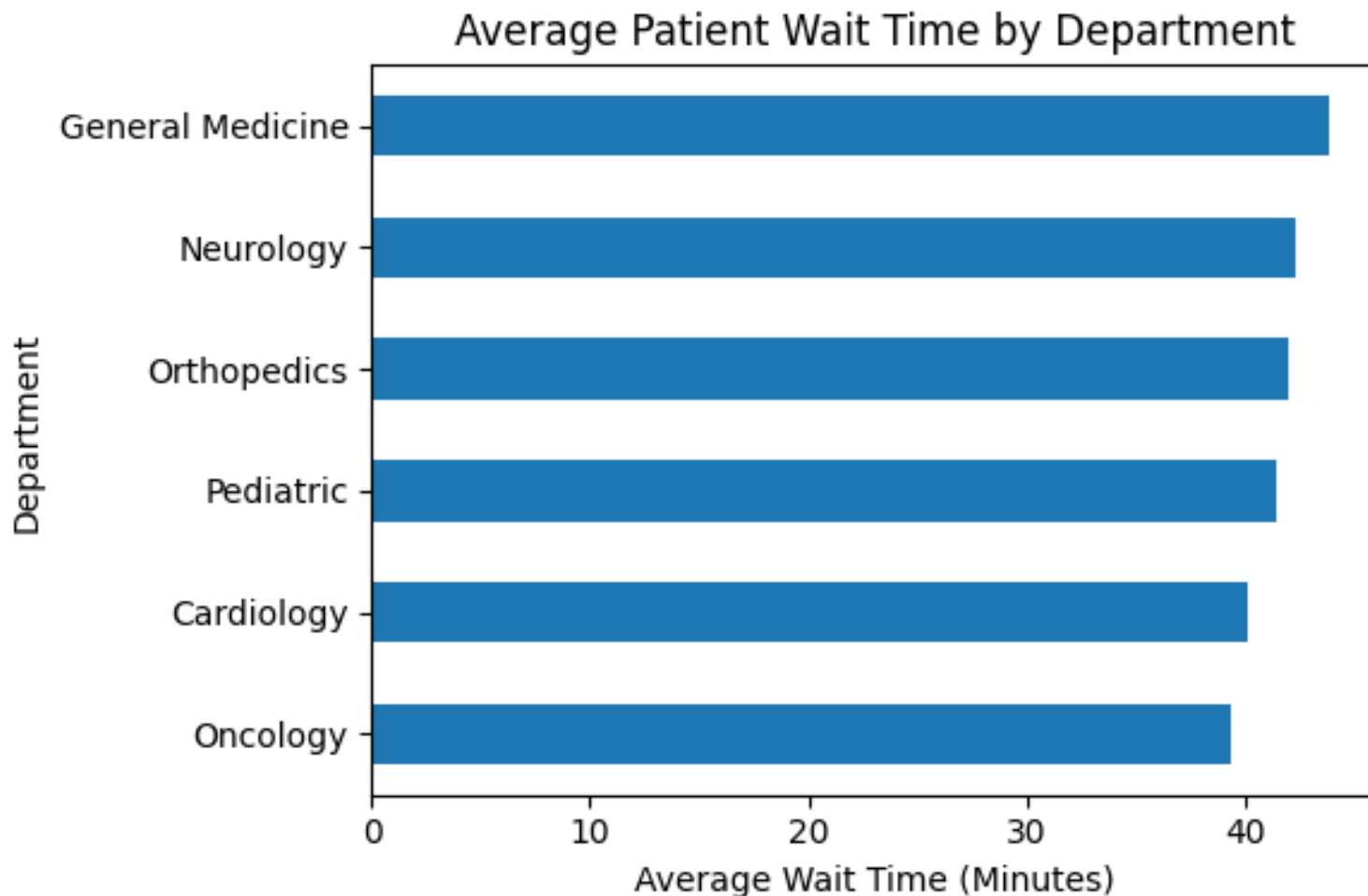
Process Mapping

- As-Is Process: Manual appointment scheduling caused delays and inconsistent updates.
- To-Be Process: Automated scheduling with real-time notifications and dashboard monitoring.
- As-Is Process: Reactive resource allocation led to department bottlenecks.
- To-Be Process: Predictive resource planning improves efficiency and availability.

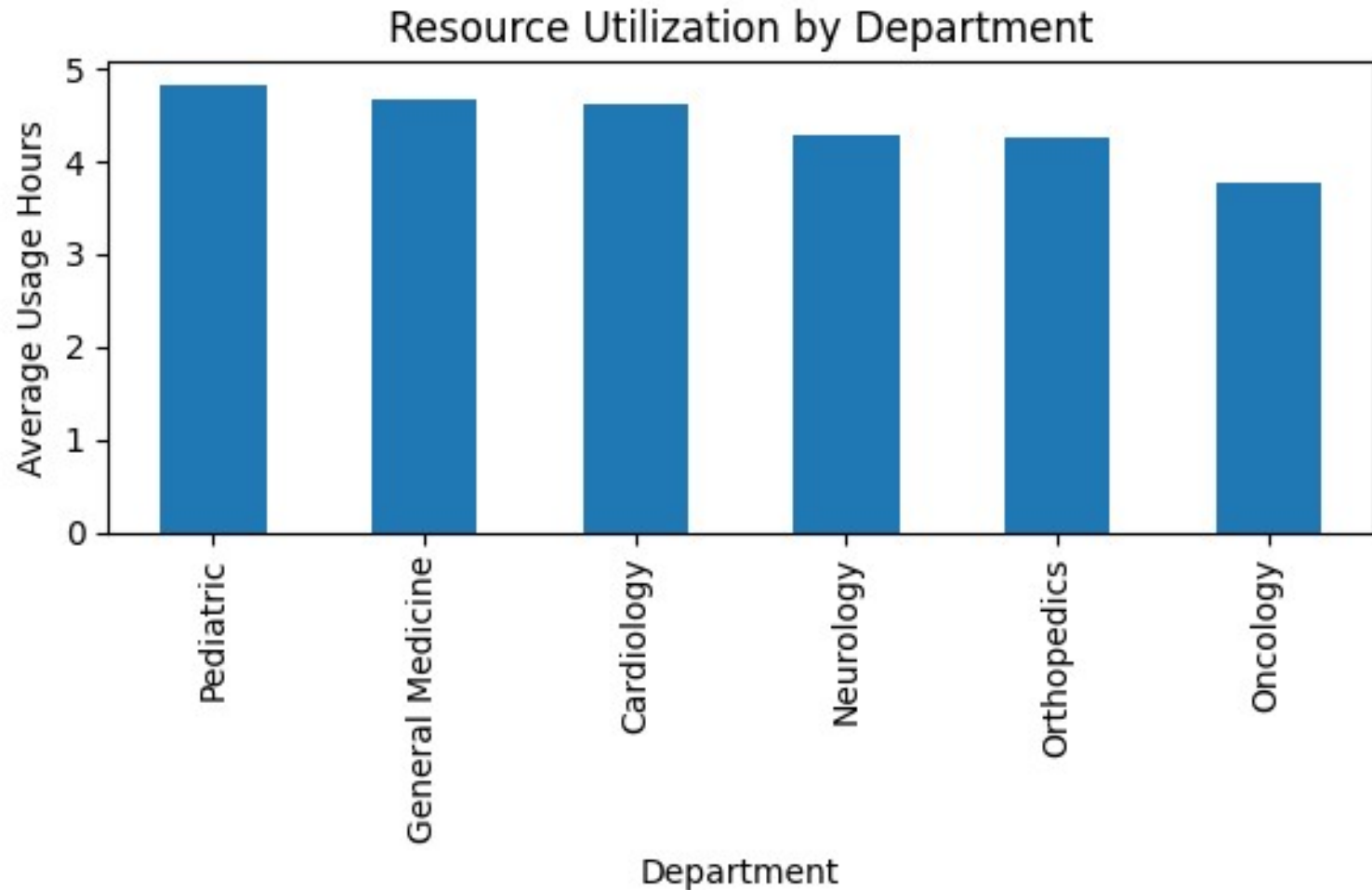
Advanced Process Mapping

- BPMN workflows clarified patient scheduling, approvals, and escalation paths.
- Swimlane diagrams defined stakeholder ownership and responsibilities.
- Process mapping reduced ambiguity in operational handoffs.
- Improved visibility supports faster issue resolution and accountability.

Data Visualization - Wait Time Analysis

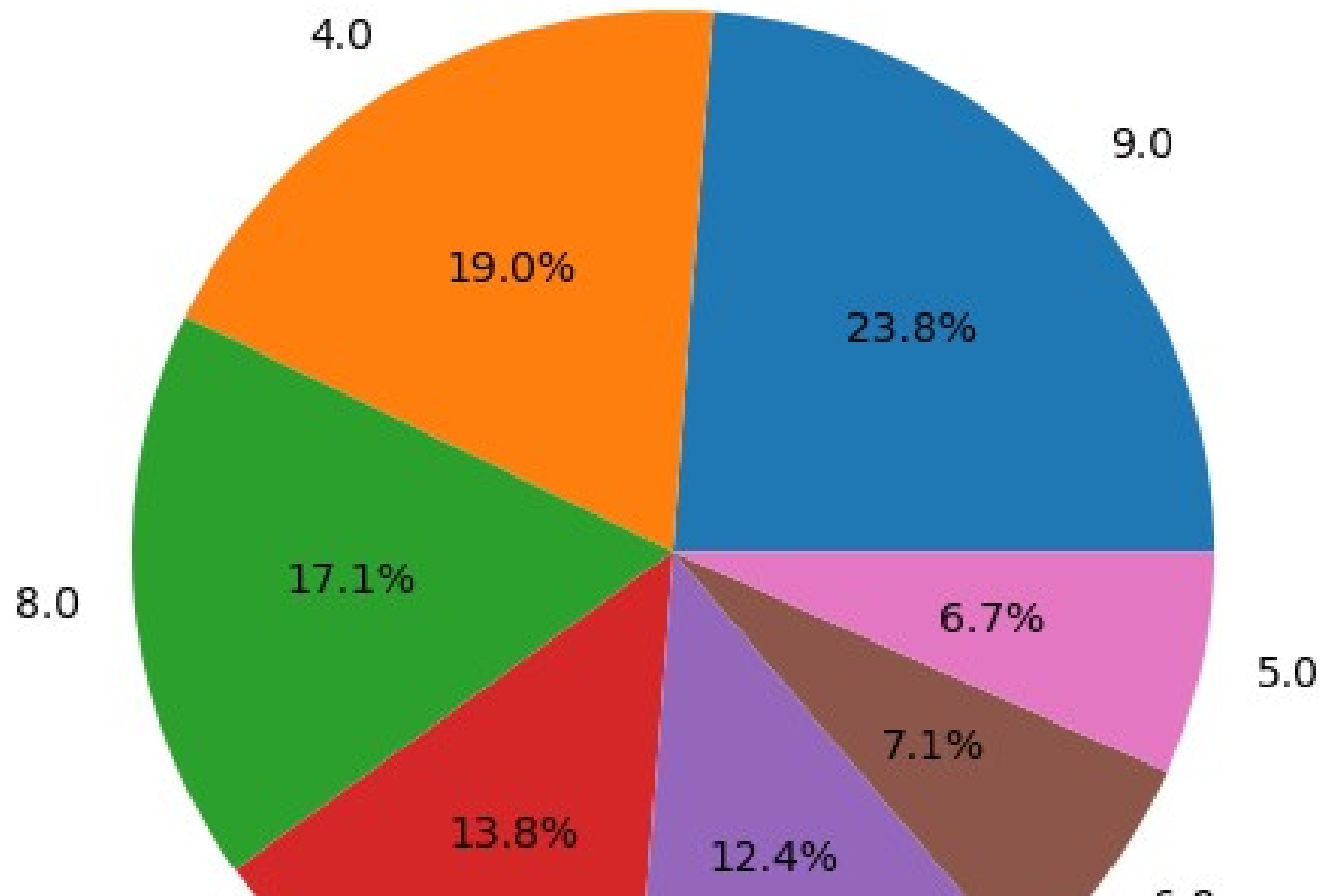


Data Visualization - Resource Utilization

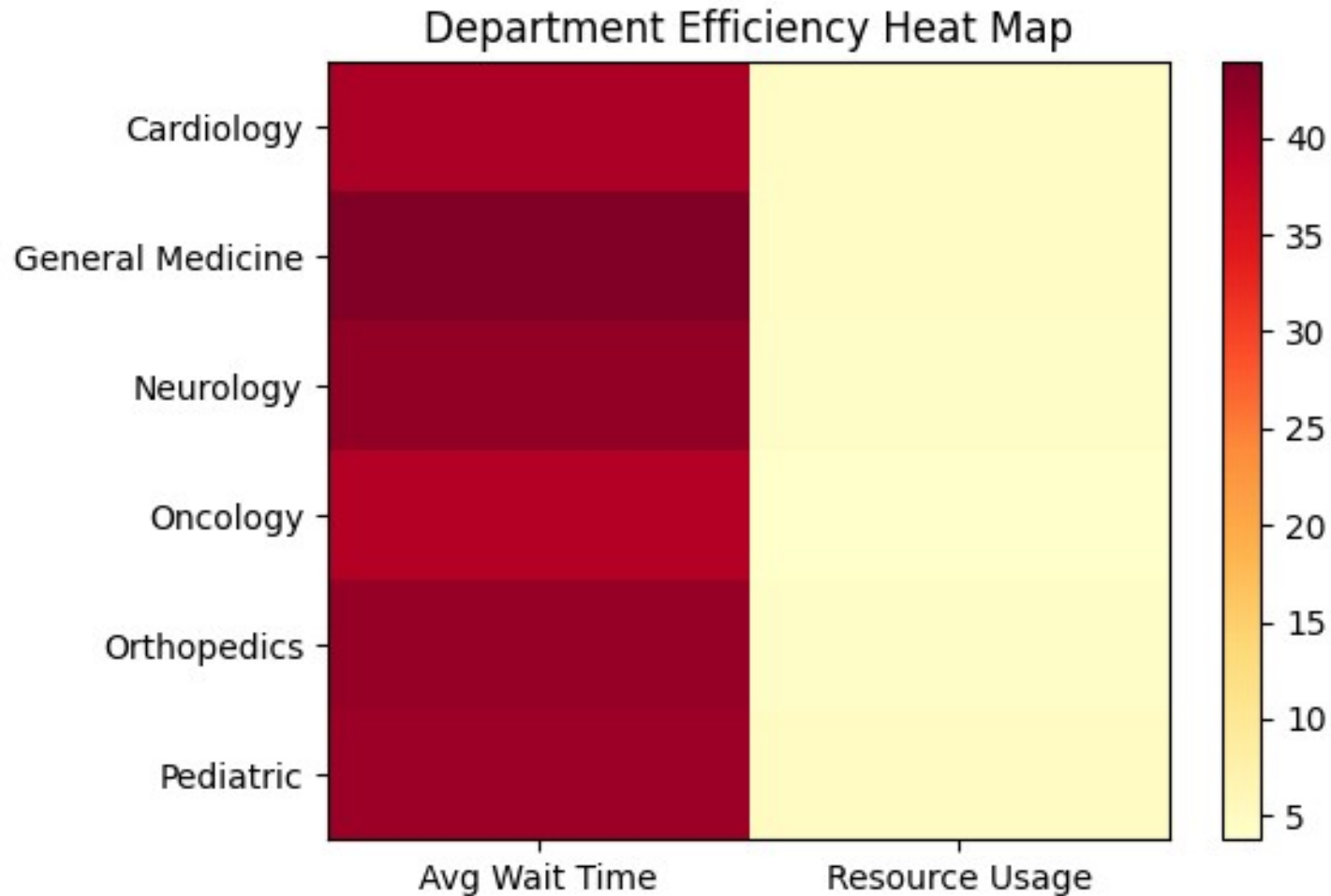


Data Visualization - Patient Feedback

Patient Feedback Distribution



Data Visualization - Department Efficiency Heat Map



Data Analysis Insights

- Average patient wait time ranged from 39.4 to 43.9 minutes across departments.
- Cardiology and General Medicine experienced the highest scheduling pressure.
- Resource utilization data identified uneven equipment and staffing usage.
- Higher wait times corresponded with lower patient satisfaction scores.
- Operational bottlenecks were concentrated during peak appointment periods.

Risk Assessment Plan

- Operational Risk: Scheduling system downtime may disrupt patient appointments.
- Technology Risk: Legacy system integration may delay implementation.
- Resource Risk: Limited staffing could impact rollout timelines.
- Compliance Risk: Patient data privacy must remain protected.
- Mitigation strategies include phased deployment and staff training.

Risk Mitigation Plan

- Implement phased deployment to reduce operational disruption.
- Establish backup scheduling procedures for downtime events.
- Conduct user training sessions before rollout.
- Monitor KPIs through dashboards and weekly governance reviews.
- Use contingency staffing plans during peak operational periods.
- Prioritize high-severity risks for immediate

Key Findings

- Long patient wait times significantly reduced satisfaction scores.
- Resource utilization patterns revealed operational inefficiencies across departments.
- Improved communication and automation can reduce bottlenecks and improve patient experience.

Key Recommendations

- Deploy centralized appointment scheduling software.
- Implement real-time operational dashboards for leadership visibility.
- Improve staff coordination through standardized communication workflows.
- Adopt predictive resource allocation and staffing models.

Conclusion

- The analysis identified operational inefficiencies impacting patient experience.
- Data-driven insights supported targeted recommendations for process optimization.
- Risk management planning improved implementation readiness.
- Operational dashboards and automation are expected to improve efficiency and satisfaction.
- The proposed solutions align with HealthFirst